Alpha Submission — README (Validation & Run)

This alpha is implemented on the **software side**. It applies a **weight-combine factor (alpha)** during **Conv + BatchNorm fusion**, producing fused weights that are then exported into HW-style text files for verification.

Overview

1. Run the `weight_combine_data` notebook to train (or load) a VGG16 variant that **inserts an 8×8 Conv layer and a BN layer**.
2. Fuse the target Conv layer and BN layer on the software side to create **new fused weights** (with alpha applied).
3. Export **activation**, **weight_kij**, and **psum** files into the `datafiles/` folder.
4. Replace the hardware input folder with `datafiles/` to run hardware verification.

- ## How to Run

  ### A) Run the notebook

  Open and run:

  - `weight_combine_data.ipynb`

  This notebook is configured for a **MacBook environment**.

  ### B) Training / loading a pretrained model

  - The notebook can train the modified VGG16 model (with an inserted 8×8 Conv layer and BN).

  ### C) Generate data files for hardware verification

  After the model is trained/loaded, run the notebook cells that:

  - fuse the target Conv layer and BN layer on the software side (alpha applied during fusion),
  - export the HW-style verification files into:
    - `datafiles/`

  The generated folder includes:

  - `activation.txt`
  - `weight_kij0.txt` … `weight_kij8.txt`
  - `psum.txt`

Outputs

After Conv+BN fusion (with alpha), the notebook generates the following files in:

- `datafiles/`

Typical contents include:

- `activation.txt`
- `weight_kij0.txt` … `weight_kij8.txt`
- `psum.txt`
- (optional) metadata such as scale/alpha info depending on the notebook

The `datafiles/` folder in this submission already contains:

- the extracted/exported HW-style files
- a PDF export of the executed notebook (for reference)

Hardware Validation

To validate in hardware:

1. Locate the hardware folder in Part 1.
2. Replace the existing hardware folder with the exported one:

- Replace:
  - `part1/hardware/datafiles/`
- With:
  - this submission's `datafiles/`

After replacement, run the existing hardware simulation flow normally using the updated datafiles.

Notes

- This alpha is **software-only** and focuses on **Conv + BN fusion** with an alpha scaling on fused weights.
- The notebook environment is **MacBook-oriented**.
- Using the provided `result/` checkpoint and `datafiles/` can reduce validation time and avoid grading delays.